

Introduction

The ability to discriminate between quantities, also known as number sense, is widely observed across animal taxa, from insects to primates. However, the capacity to perform arithmetic-like operations, such as adding or subtracting elements based on learned rules, represents a more advanced level of numerical cognition. Recent evidence suggests that certain fish species may possess this proto-arithmetic capability.

A seminal study by Schluessel et al. (2022) demonstrated that cichlids and stingrays can learn to associate abstract visual symbols with “+1” and “−1” operations. When presented with novel quantities, these fish applied the learned rules correctly, indicating the ability to manipulate numerical information according to abstract, symbolic principles. Such findings challenge traditional views that higher-order numerical cognition is restricted to mammals or birds, instead suggesting that rule-based numerical reasoning may have evolved convergently across distant taxa.

In contrast, research on zebrafish (*Danio rerio*) has predominantly focused on relative numerosity; the ability to distinguish between sets based on magnitude rather than abstract numerical rules. For example, Potrich et al. (2015) found that zebrafish can reliably discriminate between sets with favorable ratios (e.g., 2 vs. 3 or 3 vs. 4), but not finer contrasts such as 5 vs. 6. These limitations are consistent with performance governed by the Approximate Number System (ANS), a mechanism that supports approximate but not symbolic or rule-based arithmetic.

More recently, Potrich et al. (2024) attempted to test “proto-arithmetic” abilities in zebrafish by exploring whether they spontaneously preferred larger shoals, which could reflect an implicit understanding of “more.” However, because zebrafish, particularly males, are naturally motivated to join larger groups of conspecifics, such preferences may stem from social

or reproductive biases rather than genuine numerical reasoning. Consequently, current evidence does not conclusively establish whether zebrafish can engage in abstract, rule-based numerical operations akin to those demonstrated in cichlids and stingrays.

This proposal aims to address this gap by adapting the symbolic rule-learning paradigm developed by Schluessel et al. (2022) to zebrafish. Instead of addition or subtraction, our study will test whether zebrafish can learn and apply a ratio rule (1:2), a form of proportional reasoning that extends beyond linear arithmetic. In this design, zebrafish will first undergo baseline tests to rule out size, color, or stimulus preferences, followed by associative learning where color cues indicate whether the rewarded choice corresponds to a larger or smaller numerical stimulus.

By later combining shape stimuli with conspecific images and progressively removing social cues, we aim to determine whether zebrafish can generalize learned associations to abstract numerical relations. Demonstrating proportional rule learning (1:2) in zebrafish would suggest that their numerical cognition extends beyond magnitude discrimination, providing evidence for proto-multiplicative reasoning - a significant cognitive step toward abstract, symbolic thought.

References

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Methods

Ethics Information

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Pilot Data

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Design

Sampling

Apparatus and Environmental Controls

All the tests will be conducted in an experimental tank of ____ dimensions. The tank will be partitioned into two parts with an opaque divider in the middle. The first compartment will be the living area where the fishes are allowed to swim freely. (add descriptions of the living area). The second compartment will serve as the experimental chamber. The gateway from the first to second compartment will be a manually controlled guillotine door of ____ dimensions. In the experimental chamber, there will be two stimuli presented. These stimuli will be separated using an opaque divider protruding ____ cms from the wall. This will separate the decision zones for the fish (Schluessel et al., 2022).

The tank walls of the experimental chamber will be covered with black cloth to control for any environmental cues that may affect the fishes' decision. Additionally, the tank will be

rotated 90° after every quarter of the trials to further ensure that these extraneous variables do not affect the results of the study. All trials will be recorded using a camera placed over the tank.

Stimulus Design

Two dimensional shapes such as circles, squares and triangles will be used as stimuli tokens. They will be presented using laminated cards in one of two colours. Conspecific stimuli will include pictures of zebrafish which are of similar size to the shapes. All shapes within a stimuli will be of the same surface area. In Hagen et al. (2023) "*Innate colour preference in zebrafish (Danio rerio)*", the authors tested zebrafish in a plus shaped maze with four coloured zones (red, green, yellow, blue) and a central neutral zone. They found that adult zebrafish spent significantly more time in the blue zone compared to the red and yellow zones, and there was no significant difference between red and yellow zones. From this, we hypothesise that red and yellow can act as relatively neutral or non-aversion colours in the zebrafish context.

General Training Procedure

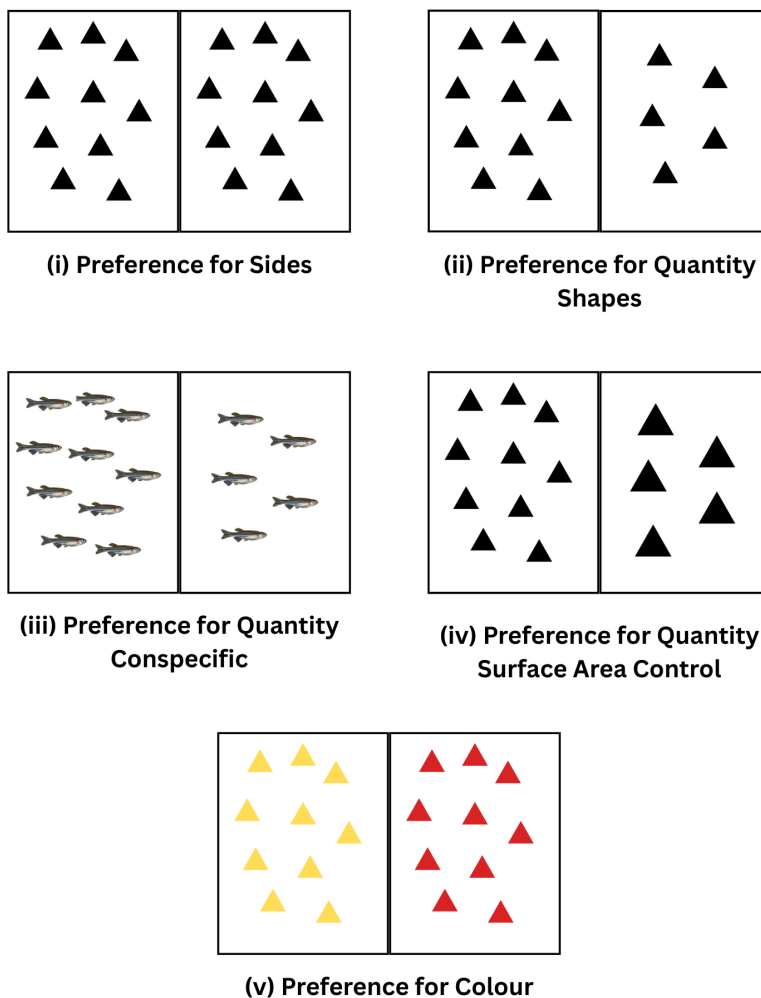
Base Line Tests

First, the fishes will be presented with the same stimuli on either side of the decision zones. This is to test if the fishes have an innate preference for a particular side or if there are any environmental confounds that may be leading the fish to a particular side. Second, the fishes will be presented with stimuli that have different numbers of shapes/fishes (conspecific stimuli) to see if they have a preference for larger or smaller quantities. Third, we will do the same but control for surface area and ensure that both stimuli have the same surface area occupied by shapes.

Fourth, we will present the same stimuli but in different colors and observe if the fishes have preference for a particular colour. These tests will help us attribute the observations to learning and manipulation of our variable of interest rather than any pre-existing biases. Here, there will be no reward or learning and the stimuli will be counterbalanced on either side.

Figure 1

Sample stimuli for baseline tests



Associative Learning

During the associative learning phase, the fish will be trained to use colour cues to determine whether the rewarded option corresponds to a larger or smaller numerical alternative. If the test stimulus is presented in yellow, the fish will be required to choose the smaller quantity, whereas if the test stimulus is red, the fish will be required to choose the larger quantity. To control for potential colour biases, this will be counterbalanced across half of the subjects. This may depend on the baseline test results for color preference in the fish.

Each trial will begin with the opening of the manual guillotine door to allow the fish to swim to the experimental chamber. Following this, two choice stimuli will be presented simultaneously in the decision zones. One stimulus will contain a higher number of items and the other will contain a lower number. The correct choice will depend on the colour of the stimuli. Training will occur over ___ sessions each day, each consisting of ___ trials. Training will continue until the fish reaches the learning criterion of at least 70% correct responses across three consecutive sessions or until the correct response is given four times in a row.

Multiple different stimuli with varying quantities and arrangements will be used to ensure that the fish is engaging in quantity discrimination and not any other kind of pattern recognition or memorisation. Correct responses will be reinforced by the delivery of a food pellet above the stimulus, in the decision zone.

Experimental Design

Following the baseline tests testing for a preference for a particular side, colour, surface area, stimuli; The subject fish is expected to associate different numerical values with different colours, stimuli and surface area.

Post this, the subject fish will be presented with an addition of two different stimuli, which will be a specified number of conspecifics alongside the same number of shape stimuli which they have been trained on in the baseline tests. The experiment will be conducted to test whether the fish can discriminate between numerical values by associating the number of shape stimuli present with the number of conspecifics present alongside for ___ number of trials.

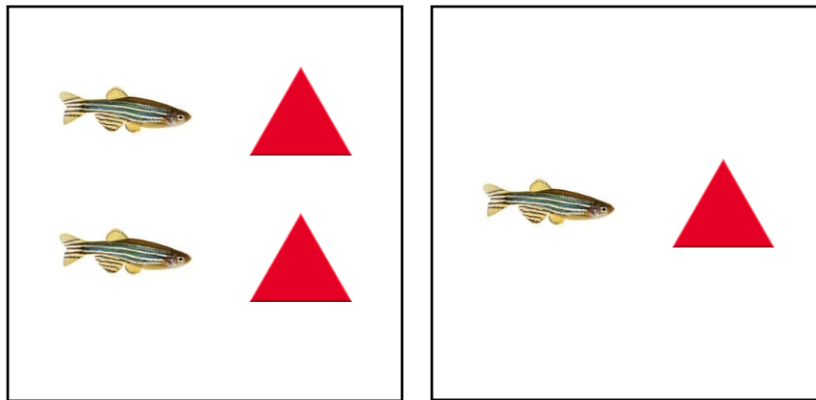
The experiment will be conducted in a rectangular gray apparatus. The tanks will be halved by a partition wall into the front and rear compartments. The apparatus will contain sand at the bottom for environmental enrichment. Using a projector or a monitor screen, pictures will be presented to the fish, containing pictures of other conspecifics and shapes. Using pipettes attached to long tubing, food pellets will be expelled immediately for every correct choice. The temperature in the tank will be maintained from 26 C- 28 C.

Trials will be conducted to teach the fish to associate the number of shapes present with the number of conspecifics with the hope that they eventually understand the number of shapes without the conspecifics present.

The experiment will commence as soon as the subject fish will cross the partition door into the choice compartment of the apparatus. In the first level of the experiment, the fish will be trained to associate the number of conspecifics with the number of shapes present alongside it. The two choices presented will be pictures of two conspecifics (zebrafish) along with two triangles, one conspecific (zebrafish) along with one triangle. The fish will be rewarded with a

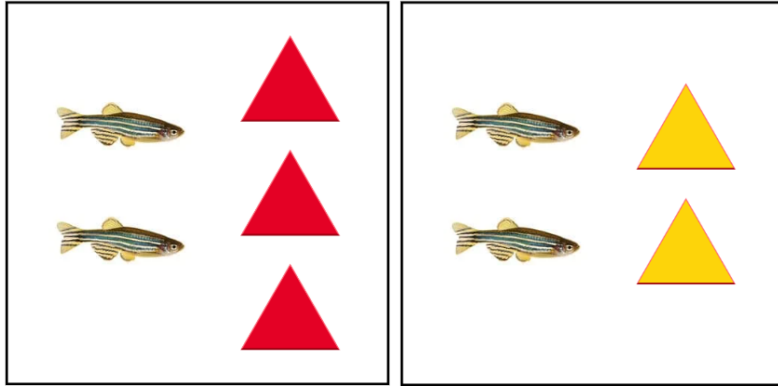
food pellet only if they choose the larger stimuli. The side of the stimuli will be switched up often as the trials continue to avoid side bias and check if the subjects are actually moving towards the larger shapes.

Figure 1



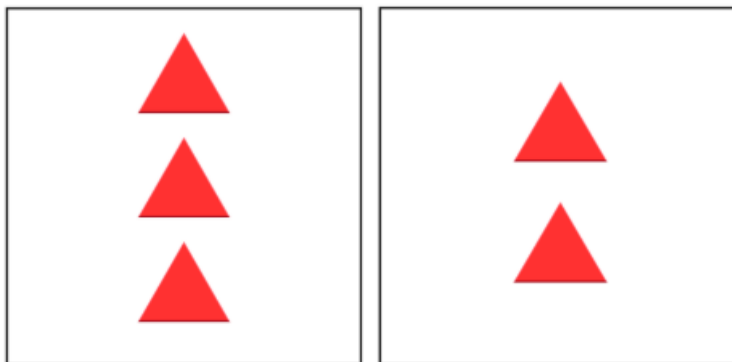
Secondly, following the associative learning tasks where the subjects are trained based on colour cues, the experiment will test their learning rates in addition to conspecifics being present alongside the shapes. For this, we will present the fish with the same number of conspecifics, however, one side will present three red triangles, whereas the other side will show two yellow triangles.

Figure 2



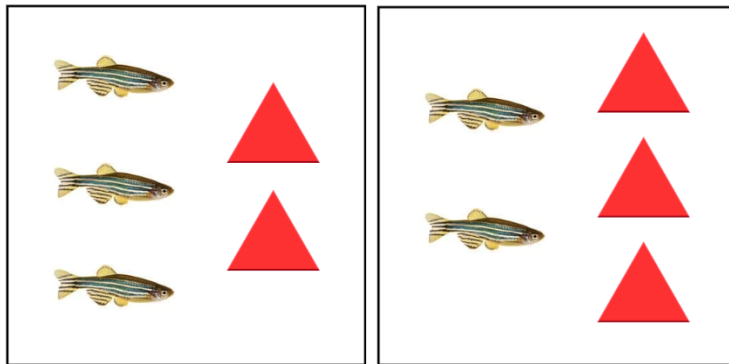
Post this, the experiment will judge if the subject can actually do numerical cognition and move towards the larger stimulus. This can be done in 2 ways- The paradigm stays the same except the conspecifics will not be presented to the subject anymore, only the triangles, out of which they will have to choose the larger one. A food pellet will be presented if they choose the larger stimuli.

Figure 3



Secondly, the conspecifics are still shown to the subject, However, the number of triangles are changed. The stimuli presented can be three conspecific zebrafishes with two triangles, and two conspecific zebrafishes with three triangles. This will be used to judge that, if after multiple trials, the subject can discriminate between the larger number of triangles and move towards that as the correct stimuli.

Figure 4



Trials will continue until the subject fish gives the correct response at least 5 times in a row for each of the designs. Multiple stimuli with different quantities will be used to observe if the fish can learn to associate and discriminate between quantities through the help of conspecifics.

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Expected Results

It is expected that the zebrafish will demonstrate the ability to discriminate between quantities and learn colour-quantity association through reinforcement. During the associative learning phase, zebrafish trained to associate red with larger numerosity and yellow with smaller numerosity (or vice-versa) are expected to reach the learning criterion of $\geq 70\%$ correct choices across three consecutive trials. This is consistent with prior findings which suggest that zebrafish can acquire numerical rules under reward-based paradigms (Bisazza & Santacà, 2022).

The null hypothesis, that zebrafish will perform at chance level (0.5 accuracy) and fail to learn the association, will be rejected if the mean accuracy is significantly greater than 0.5, determined through one sample t-test with $p < 0.05$.

Across base trials the fish are expected to show no systematic bias towards side, colour or surface area. This confirms that learning reflects cognitive discrimination rather than innate preference (Roy et al., 2019). Choice accuracy is expected to increase progressively across sessions with shorter latency to correct responses, indicating reinforcement-based learning during the associative phase. This essentially means that the fish are not relying on perceptual queues but on numerical difference between stimuli (Bisazza & Santacà, 2022).

In the next test phase involving conspecifics and shapes, zebrafish are expected to prefer displays representing larger total numerosity (3 fish + 3 triangles vs. 2 fish + 2 triangles). This suggests the ability to integrate numerical information across stimuli categories (Bisazza & Santacà, 2022). When colours are introduced in the presence of equal conspecifics (3 red

triangles vs. 2 yellow triangles) they are expected to follow the learned rule rather than being guided by simple visual salience (Bisazza & Santacà, 2022). Finally, when the conspecifics are removed zebrafish are expected to maintain performance above chance, indicating transfer of learning. Following Weber's law, accuracy is supposed to decrease as numerical ratio approaches unity (Messina et al., 2021).

Rationale for non-significant results chatgp

Several factors can cause the results to fail to reach statistical significance. Zebrafish may exhibit colour bias. Previous studies report that preference for specific hues like red, can influence learning and hinder reversal or rule learning (Roy et al., 2019). Furthermore, individual variation in stress, temperature, motivation as well as environmental factors may reduce responsiveness. Failure to discriminate may arise if fish rely on continuous cues like surface area rather than numerosity, as observed in other shoaling studies (Bisazza & Santacà, 2022). Lastly, suboptimal reinforcement timing or insufficient training trials could prevent rule learning.

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Discussion

The zebrafish showed no significant preference for side, shape, conspecific, surface area or colour, which allowed us to control these factors in our experiment. In phase 2, zebrafish were able to learn to associate colours with greater or smaller numerosities. They were able to successfully grasp this even with mixed sets of conspecifics and shapes, as well as transfer this association when triangle numbers were unequal, or conspecifics were removed. This result is similar to the study by Potrich et al. (2015), which saw zebrafish successfully identify the larger numerosity until the ratio of 2:3, at which point they failed to differentiate, regardless of conspecific and triangle combination.

In line with previous literature (Potrich et al., 2015; Bisazza & Santacà, 2022), the zebrafish showed decreasing accuracy with larger numbers in each set. The fish did show greater accuracy at large sets if the ratio is smaller than small sets with a small ratio, implying a stronger ratio effect. This is also consistent with the quantity discrimination study by Potrich et al. (2015). If the fish fail to meet these criteria, a possible reason may be due to a preference found in the baseline tests. We may also discover that zebrafish have a limited ability to discriminate between quantities, which ratio and set numerosity may play a role in.

However, there is also the possibility of the fish relying on other spatial cues to navigate the paradigm rather than just numerical abilities. The study, like most studies, was limited to just visual stimuli whereas previous research with mammals and birds also demonstrates numerical discrimination through auditory and tactile stimuli (Agrillo et al., 2017). This suggests the existence of varying numerical systems that were out of the scope of investigation.

The findings have significant implications for understanding fish behaviour from using numerical cues to make informed ecological decisions including choice of shoals to join, sites to

forage, and protection from predators (Salena et al., 2021). Fish often show preference for larger groups as shown by a study conducted by Dadda et al. (2009) where barriers allowed them to see only one fish at a time suggesting the importance of numerical cognition in crucial life processes.

References

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